CY4001: Industrial Chemistry

Unit 3:Fuel Chemistry

Unit 3: Fuel Chemistry

Gaseous Fuels

- Gaseous fuels occur in nature, besides being manufactured from solid and liquid fuels.
- Common gaseous fuels are liquefied petroleum gases (LPG), natural gas, producer gas, blast furnace gas, and coke-oven gas.

Advantages

- They can be conveyed easily through pipelines to the actual place of need, thereby eliminating manual labor in transportation.
- They have high heat contents and hence help us in having higher temperatures.
- Their combustion can readily be controlled for changes in demand like oxidizing or reducing atmosphere, length of flame, temperature, etc.
- They can clean in use and do not require any special burner.
- They burn without any soot, or smoke and ashes.
- They are free from impurities found in solid and liquid fuels.

Disadvantages

- Very large storage tanks are needed.

They are highly inflammable, so chances of fire hazards in their use are high.

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Natural Gas

- Natural gas is a naturally occurring hydrocarbon gas mixture consisting of methane and commonly including varying amounts of other higher alkanes.

The composition of natural gas is given below.

Constituent	Percentage
Methane	88
Ethane	06
Propane	04
Butane	1–1.5
Pentane	0–0.5

The calorific value is 12,000–14,000 kcal/m^3.

Uses

- It is used as a green fuel for engines, preferably in highly populated cities and areas having dense vehicle population.

- In India, at New Delhi, all the public transportation vehicles are fitted with engines which use CNG (Compressed Natural Gas) as the fuel.

- It is an excellent domestic and industrial fuel and can be conveyed over very large distances in pipelines.

Water Gas or Blue Gas

- It is essentially a mixture of combustible gases, CO and H_2 along with a little non combustible gases like CO_2 and N_2.
- It is also known as blue gas because it burns with a blue flame due to combustion of CO.
- The calorific value of water gas is about 2800 kcal/m^3.
- The average composition of water gas is as follows: CO: 40−45%; H_2: 45−50%; CO_2:4%; N_2:4%.

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Water Gas Manufacture:

- **It is produced by passing steam and little air alternatively through a bed of red hot coke maintained at 1000°C.**

Principle: When steam is blown through a bed of hot coke (1000°C) water gas is produced.

$$(C + H_2O \text{ steam}) \rightarrow CO + H_2 \qquad \Delta H = 28 \text{ kcal}$$

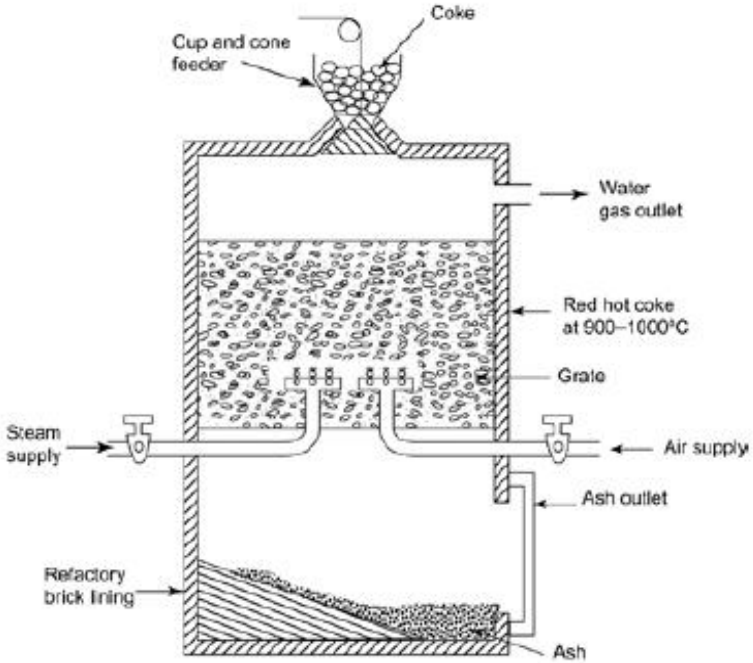
- The reaction being endothermic in nature, the temperature of the coke bed gradually decreases with continuous passage of steam and the drop in temperature must be prevented.

- For this the steam supply is temporarily cut off and air is blown in. The overall reaction during air blow is the formation of CO.

$$2C + O_2 \rightarrow 2CO \qquad \Delta H = -19 \text{ kcal}$$

- The reaction being exothermic increases the temperature of the coke bed to about 1000°C.

- Thus by blowing steam and air alternatively, the temperature of the coke bed can be maintained at 1000°C.



Water Gas

Uses:

- **Enriched water gas (mixed with hydrocarbons), which burns with luminous flame is used as illuminating agent and heating purposes.**
- **Water gas is also used for welding purposes and as a fuel in glass and ceramic industries.**
- **It is used for the manufacture of methyl alcohol and synthetic petrol.**

Producer Gas

- It is essentially a mixture of CO and N_2.
- Prepared by passing air mixed with little steam over a red hot coal bed maintained at about 1100°C.
- The average composition of producer gas is as follows: CO: 25-30%; N_2: 50−55%; H_2: 10%; CO_2: 5%; Hydrocarbons: 2-3%
- The calorific value of producer gas is 1800 kcal/m^3.

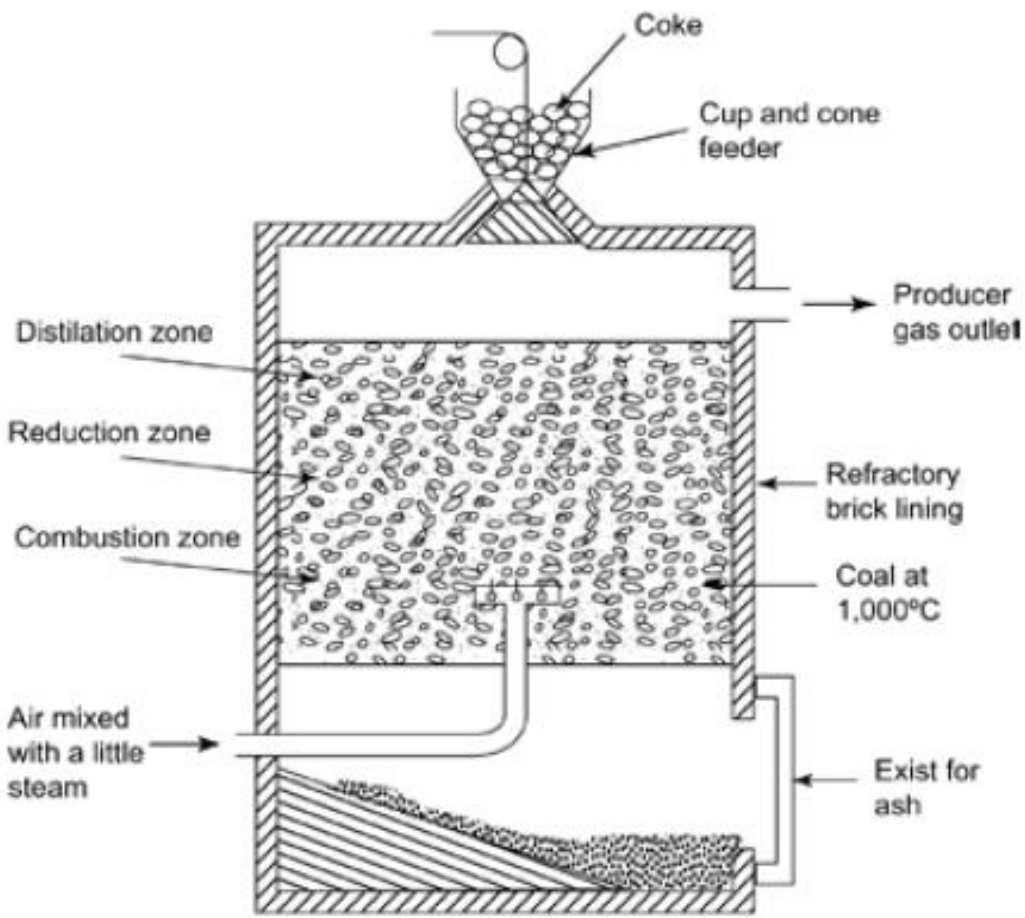
Properties:

- It is a poisonous gas.
- It is insoluble in water.
- It is heavier than air.

Manufacture:

- The furnace used for the manufacture of producer gas is known as producer.

Combustion Zone: When a mixture of air and little steam is passed through a bed of red hot coal, carbon (of the coal) combines with oxygen (of the air) in the lower part of the furnace to form CO_2. [$C + O_2 \rightarrow CO_2$ $\Delta H = 97$ kcal]



Gas producer

Reduction Zone: CO_2, so formed, rises up through the red-hot coal and gets reduce to CO during its passage [$CO_2 + C \rightarrow 2CO$ $\Delta H = -39$ kcal].

Distillation Zone: This zone is the topmost in the coke bed. The incoming coke is heated by the outgoing hot producer gas which distillates the volatile matter present in the coke.

Uses:

- It is used as a fuel in the manufacture of steel, glass, coal gas.
- It is used as a reducing agent in metallurgical operations.

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Liquefied Petroleum Gas (LPG)

- LPG is defined as those hydrocarbons, which are gaseous at normal atmospheric pressure, but may be condensed to the liquid state at normal temperature, by the application of moderate pressures.

- Although they are normally used as gases, they are stored and transported as liquids under pressure for convenience and ease of handling.

- Liquid LPG on evaporation produces 250 times more volume of gas.

Composition of LPG

- **n-butane = 70 %**
- **Isobutane = 17 %**
- **n-propane = 11 %**
- **Butylene and ethane = rest**

Uses:

- **Used as a fuel for domestic cooking.**
- **Used for heating industrial furnaces.**
- **Used as an alternate for gasoline in automobiles.**

Calorific Value:

- **Its calorific value is 27,000 kcal/m^3.**

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Coal Gas

- **It is a mixture of H_2 and CH_4.**
- **Its calorific value is 5000 kcal/m^3.**
- **Its approximate composition is 50% H_2 + 30% CH_4 + 7% CO + the rest N_2, CO_2 and alkenes.**

Manufacture:

$$\text{Coal} \xrightarrow[\text{Absence of air}]{1200^\circ C} \text{Coke} + \text{Coal gas}$$

Uses:

- **Coal gas is used as a domestic fuel. It is used for maintaining a reducing atmosphere in metallurgy.**
- **It is also used in smelting of metals and alloys.**

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Biogas

- Biogas typically refers to a gas produced by the biological breakdown of organic matter in the absence of oxygen.
- Biogas originates from biogenic material and is a type of biofuel.
- The cheapest and easily obtainable biogas is gobar gas (biogas produced from cow dung), which is produced by anaerobic fermentation of cattle dung.
- The main constituent present in gobar gas is methane.
- The calorific value is 5400 kcal/m^3.

The general composition of biogas is

Compound	%
Methane	50–75
Carbon dioxide	25–50
Nitrogen	0–10
Hydrogen	0–1
Hydrogen sulphide	0–3
Oxygen	0–2

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Oxygenated fuel

Oxygenated Gasoline	Non-Oxygenated Gasoline
Oxygenated gasoline is a type of gasoline in which oxygenated chemical compounds are added to increase its oxygen content.	Non-oxygenated gasoline is the pure form of gasoline obtained from the distillation process of crude oil.
Oxygenated gasoline additives make it more combustible but it reduces its fuel mileage and performance.	Non-oxygenated gasoline burns slowly as compared to oxygenated gasoline and gives more fuel mileage and performance.
Oxygenated gasoline has additives like ethanol, ethers, butanol, etc.	Non-oxygenated gasoline doesn't have any additives. It is pure gasoline made of hydrocarbons.
Oxygenated gasoline has lower carbon monoxide and soot (PAH) emission levels.	Non-oxygenated gasoline has higher carbon monoxide and soot (PAH) emission levels.
Oxygenated gasoline is used for passenger vehicles and light-duty trucks.	Non-oxygenated gasoline is used for older boats, lawnmowers, and sports cars also.

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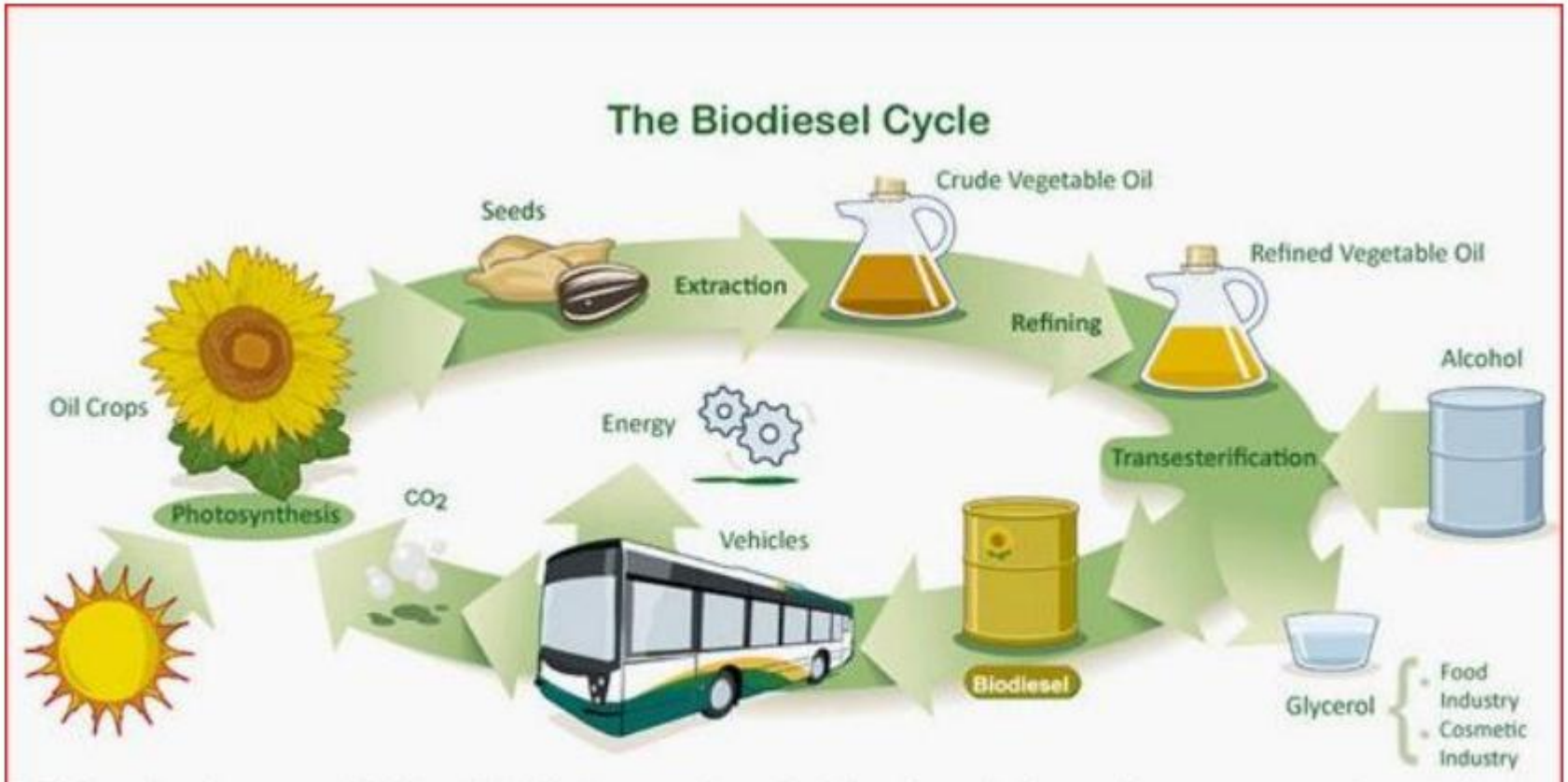
Biodiesel

Biodiesel is a renewable, biodegradable, and clean-burning fuel that can be used as a substitute for petroleum diesel. It's made from biological sources like animal fats, vegetable oils, or recycled grease



Biodiesel releases only the CO2 that was absorbed by the plants as they were growing and making oil

Advantages of Biodiesel

- Renewable fuel, obtained from vegetable oils or animal fats
- Low toxicity, in comparison with diesel fuel.
- It is the only alternative fuel that can be used in a conventional diesel engine, without modifications.
- It can be blended with diesel fuel at any proportion
- Used cooking oils and fat residues from meat processing may be used as raw materials.

Disadvantages of Biodiesel

- Slightly higher fuel consumption due to the lower calorific value of biodiesel.
- It is less stable than diesel fuel, and therefore long-term storage (more than six months) of biodiesel is not recommended.

Oil Crops Useful for Biodiesel Production

Various vegetable oils like sunflower, soybean, rapeseed (mustard), palm, peanut, cottonseed, corn (maize), canola, ricebran, jojoba, flax oils provide the fatty acids part in biodiesel

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Comparison of Solid, liquid and gaseous fuels

Table 1.1 Comparison of solid, liquid and gaseous fuel

S.No.	Solid Fuel	Liquid Fuel	Gaseous Fuel
1	Cheap and easily available	Costlier than solid fuel except in the countries of origin	Costly, because except natural gas all other gaseous fuels are derived from solid and liquid fuels
2	Convenient to store without any risk of spontaneous explosion	Great care must to be taken to store them in closed containers	Very large storage tanks are needed. Storing gaseous fuel requires extra care as they are highly inflammable
3	Large space is required	Storage space is less compared with solid and gaseous fuels	They must be stored in leak proof containers
4	They are easy to transport	They can be easily transported through pipelines	They can also be transported through pipelines
5	They posses moderate ignition temperature. Combustion is slow but it cannot be controlled easily	Combustion takes place readily and can easily be controlled or stopped by reducing or stopping the fuel supply	Combustion is fast and can be controlled and stopped easily

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Comparison of Solid, liquid and gaseous fuels

S.No.	Solid Fuel	Liquid Fuel	Gaseous Fuel
6	Ash is produced and its disposal is a big problem. Smoke is also produced	Ash is not produced, however fuels with high carbon and aromatic contents may produce smoke	Neither ash nor smoke is produced
7	They cannot be used in internal combustion engine	Used in internal combustion engine (petrol, diesel)	Used in internal combustion engines (CNG, LPG)
8	They have low thermal efficiency	Their thermal efficiency is higher than solid fuels	Their thermal efficiency is the highest
9	Their calorific value is lowest	Their calorific value is higher than solid fuels	Their calorific value is the highest
10	Least risk of fire hazards	Risk of fire hazards is high	Highest risk of fire hazards

Life gives you a set of cards

Just focus on your set of cards

If you play your cards right

Things will happen in the long run

In the short run.

It is anybody's guess